

RIO BRANCO, Brazil

WMO: 829170

Lat: 9.869S

Lon: 67.894W

Elev: 634

StdP: 14.36

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	58.6	61.2	48.4	51.5	68.9	53.4	62.0	70.6	13.9	74.1	12.0	74.4	5.0	180	0.262

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	22.0	97.0	77.2	95.2	76.9	93.5	77.0	83.6	88.1	82.6	87.1	81.7	86.8	6.2	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.6	174.2	85.8	81.5	167.6	85.0	80.6	162.9	84.4	48.3	88.9	47.2	87.1	46.1	87.1	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.4	11.5	9.8	DB	53.8	99.7	4.0	2.4	50.9	101.4	48.5	102.9	46.3	104.2	43.4	106.0
			WB	51.3	84.4	4.6	2.4	48.0	86.1	45.3	87.6	42.7	88.9	39.4	90.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	79.7	79.8	79.8	80.0	79.8	78.3	77.3	77.5	79.9	81.5	81.5	80.7	79.9
	DBStd	3.93	2.62	2.43	2.38	3.25	4.27	4.75	5.06	4.63	4.33	3.37	3.27	2.53
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	7	0	0	0	0	1	3	0	0	0	0	0	0
	CDD50	10829	924	833	931	895	876	818	854	928	945	977	921	927
	CDD65	5363	459	413	466	446	412	371	392	464	495	512	471	462
	CDH74	45749	3375	2991	3439	3473	3219	3069	3742	4858	5163	4774	4152	3494
	CDH80	19401	1151	985	1175	1307	1227	1306	1804	2553	2679	2234	1742	1238
(14) Wind	WSAvg	3.4	3.6	3.5	3.2	3.0	3.0	2.9	3.1	3.5	3.8	3.7	3.7	3.7
(15) Precipitation	PrecAvg	71.0	10.7	9.9	10.2	6.7	3.5	1.7	0.9	1.8	3.4	5.7	7.3	8.9
	PrecMax	101.1	20.1	14.6	15.8	16.4	8.2	6.6	2.3	4.4	6.9	9.5	15.2	14.3
	PrecMin	48.6	4.7	5.8	4.9	1.0	0.7	0.1	0.0	0.0	0.9	2.7	3.3	3.5
	PrecStd	11.8	3.9	2.2	3.0	3.1	1.8	1.5	0.7	1.1	1.7	2.0	2.7	2.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.2	92.2	92.5	93.4	93.2	93.5	96.4	100.4	100.3	98.6	95.7	93.2
		MCWB	81.6	79.7	79.9	80.9	80.6	78.5	76.7	74.2	76.8	78.5	79.9	80.2
	2%	DB	91.0	89.9	90.0	91.2	91.0	91.2	93.4	96.6	97.0	95.4	93.3	91.0
		MCWB	80.1	79.0	79.0	80.2	79.4	77.5	75.1	75.8	76.9	78.2	79.5	79.4
	5%	DB	88.8	88.1	88.3	89.4	89.2	89.4	91.5	94.6	95.0	93.3	91.3	89.2
		MCWB	79.3	78.7	79.0	79.5	78.5	77.1	75.1	74.9	76.4	78.0	78.9	79.0
	10%	DB	86.3	86.1	86.4	87.5	87.0	87.7	89.6	92.5	93.0	91.1	89.4	86.4
		MCWB	78.6	78.4	78.8	78.9	77.9	76.5	74.5	74.6	76.1	77.8	78.4	78.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	84.6	83.3	83.5	85.0	83.6	81.8	81.0	81.3	82.6	83.1	83.9	84.2
		MCDB	89.4	86.7	86.6	88.4	88.3	87.4	88.0	89.7	90.5	90.3	89.7	88.0
	2%	WB	82.5	81.6	81.9	82.7	81.9	80.4	79.2	79.4	80.4	81.5	82.3	82.5
		MCDB	86.4	85.6	85.7	86.8	87.0	86.9	87.5	89.0	89.1	88.8	87.5	86.1
	5%	WB	80.9	80.4	80.6	81.2	80.3	79.0	78.0	78.0	79.0	80.3	80.9	81.0
		MCDB	85.4	85.2	85.7	86.4	86.1	86.1	86.8	88.4	89.2	88.0	87.0	85.3
	10%	WB	79.5	79.1	79.3	79.8	78.7	77.8	76.7	76.8	77.9	79.2	79.7	79.6
		MCDB	84.3	84.1	84.5	85.2	84.5	85.1	85.5	87.2	88.3	86.7	85.9	84.1
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	13.8	14.4	15.5	16.3	18.7	22.1	23.8	22.0	18.8	16.6	14.7
		MCDBR	16.9	16.4	16.7	18.1	19.0	21.0	24.2	26.0	24.3	22.1	19.6	17.6
		MCWBR	7.6	7.3	7.4	8.5	8.5	9.3	8.7	8.1	7.5	7.5	7.7	8.0
	5% WB	MCDBR	15.3	15.0	15.2	16.4	17.6	19.6	21.3	22.9	21.9	19.4	17.5	15.3
		MCWBR	8.9	8.5	8.5	9.4	9.4	10.0	9.5	8.7	8.5	8.8	9.2	9.1
(40) Clear-Sky Solar Irradiance	taub		0.396	0.392	0.390	0.384	0.387	0.383	0.394	0.539	0.636	0.491	0.437	0.409
	taud		2.497	2.512	2.530	2.514	2.487	2.505	2.439	1.993	1.760	2.183	2.343	2.448
	Ebn at Noon		300	299	295	287	277	273	272	240	226	268	286	295
	Edh at Noon		36	36	35	34	33	31	34	56	74	49	42	38
	RadAvg		1434	1376	1427	1447	1384	1403	1550	1586	1575	1593	1548	1429
(45) All-Sky Solar Radiation	RadStd		76	62	59	93	90	98	122	112	135	80	86	104

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-2.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page